

B. Tech 1st Semester Sessional Examination-I

Subject-Mathematics-I

Branch- CE (Sec-C), Total Marks-30, Date-13.09.2024

Answer the following questions:

$$f(c) = 0$$

$$\frac{f(b) - f(a)}{b - a} \quad \times$$

1. State Rolle's Theorem. Verify Rolle's theorem for $f(x) = (x - a)^m(x - b)^n$ in $[a, b]$, where m, n are positive integers. 3+7=10

2. Find nth derivative of $(ax + b)^m$ where m is any real number 3

3. Evaluate: $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$ 2

4. Express the matrix $A = \begin{bmatrix} 1 & -1 & 2 \\ 6 & -2 & 3 \\ 3 & -2 & 4 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrices. 4+4=8

5. Reduce the following matrix to Echelon form $A = \begin{bmatrix} 3 & 4 & 1 & 6 \\ 2 & -1 & 2 & -5 \\ 1 & 3 & -1 & 9 \end{bmatrix}$ 7